

High Transmittance OLED for Under Display Face Recognition

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Abstract

This paper presents a high-transmittance OLED display for Under-Display Face Recognition (UDFR) using a novel LTPO pixel circuit with high-mobility, stable MOS TFTs. Combined with transparent wiring, a specialized COE layer, and cathode patterning, the design achieves an infrared transmittance of 29.64% in a 6.0-inch FHD prototype. This work enables seamless integration of facial recognition under the screen.

Author Keywords

Under Display Face Recognition; LTPO; COE;

1. Introduction

Since its inception in 2016, the full-display smartphone concept has significantly enhanced device aesthetics through its high screen-to-body ratio, establishing itself as the dominant trend in the industry. The core design principle involves minimizing non-display areas, particularly the "forehead," "chin," and side bezels, to maximize the active display area within a given form factor, thereby creating an immersive user experience.

While the "chin" has been effectively eliminated and side bezels reduced to under 1 mm, further minimization is increasingly challenging and offers diminishing returns for the overall screen-to-body ratio. Consequently, manufacturers have shifted focus to the "forehead," which houses essential sensors that occupy significant display space.

To reduce the footprint of the front-facing camera, solutions like pop-up mechanisms have been introduced. Although they preserve screen integrity, they introduce drawbacks such as increased mechanical complexity, added weight and thickness, and compromised dust and water resistance.

The emergence of full-display with camera (FDC) or under-display camera (UDC) technology [1-2] offers a promising alternative. This technology creates a transparent zone within the

display to conceal the camera underneath, significantly boosting the screen-to-body ratio without sacrificing display continuity. However, as this transparent area must still display images, it contains numerous opaque elements. These cause severe diffraction, significantly degrading front camera image quality. While image restoration algorithms can partially mitigate these effects, they fall short of the quality achieved by conventional front cameras and cannot fully resolve real-time diffraction issues, such as those during video calls.

Consequently, FDC phones have not yet become mainstream, with punch-hole displays remaining prevalent. For phones without Face recognition, a single front-camera hole suffices. However, for Face recognition-equipped models, either three separate holes or an extended notch (e.g., the triple-hole design of Huawei Mate 70 or Apple iPhone 17's Dynamic Island) is required. These intrusions occupy considerable display area and disrupt visual uniformity and aesthetics.

This raises a critical question: can Face recognition sensors be integrated under the display, similar to the FDC approach? While diffraction would also affect Face recognition accuracy, consumer sensitivity to this is lower compared to the readily perceptible quality loss in selfies and video calls from FDC technology. Therefore, we posit that under-display face Recognition (UDFR) represents the next frontier for smartphone innovation, as illustrated in Fig. 1.

2. Analysis of UDFR

UDFR presents distinct challenges compared to FDC technology. First, regarding the number and type of high-transmittance areas, FDC requires only one area for incident light. In contrast, UDFR necessitates at least two: one for incident IR light and another for emergent IR light (Fig. 2). Second, the light transmission requirements differ: FDC transmits visible light, while UDFR relies on infrared (IR) light.

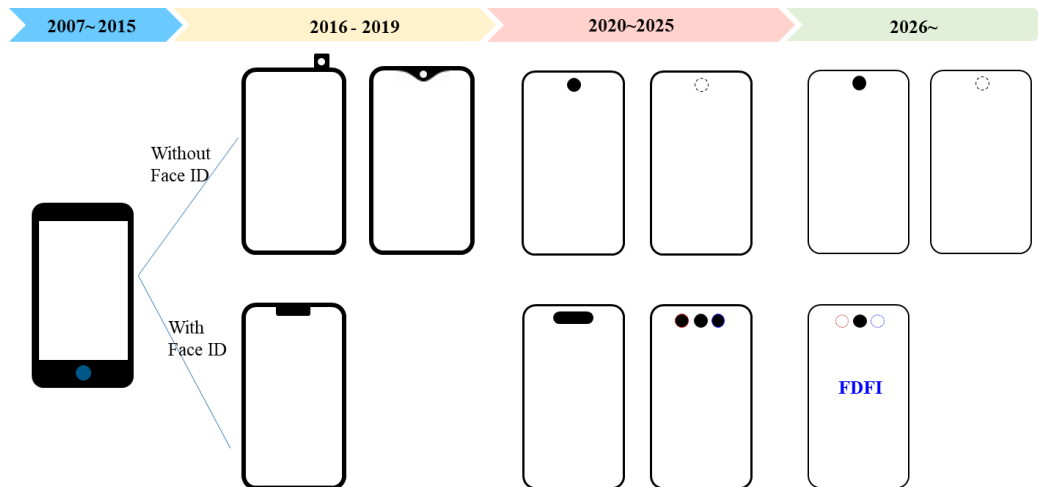


Figure 1. The changing trends of the appearance of mobile phone screens.

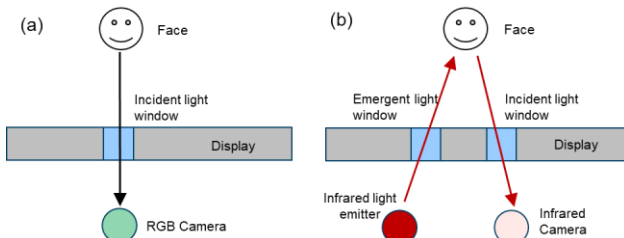


Figure 2. Optical path schematic for under-display sensor technologies: (a) FDC, (b) UDFR.

A typical OLED stack comprises a polyimide substrate, a driving circuit layer, an OLED light-emitting layer (including anodes), a cathode, a thin-film encapsulation layer, and an anti-reflection layer. Among these, the pixel circuit layer most significantly impacts IR transmittance, followed by the cathode layer.

Current strategies to mitigate transmittance loss from pixel circuits fall into two categories: pixel circuit edge-placement [3-4] and in-situ placement [5].

The edge-placement method relocates the pixel circuits from the high-transmittance area to the peripheral transition zone. This can be "one-drive-one" (one circuit per pixel) or "one-drive-multi" (one circuit drives multiple pixels). The "one-drive-multi" approach requires fewer circuits, alleviates layout pressure, and allows for a larger transparent area but suffers from inferior image quality. The "one-drive-one" method offers better display performance but demands a larger transition zone, more photomasks, and limits the high-transmittance area size.

The in-situ placement method keeps the pixel circuits in place. One sub-type reduces the PPI (pixels per inch) in the high-transmittance area, often to one-quarter of the main display, resulting in visible graininess (e.g., Samsung Galaxy Fold series). Another sub-type maintains the same PPI but compresses the circuit area to about half its original size to enhance transmittance. This still creates a visible difference from the main display and is incompatible with LTPO pixel architectures.

Given the limitations of existing methods, we propose a third, novel approach to in-situ pixel circuit design for UDFR.

3. Methods and Results

3.1 New LTPO Pixel Circuit

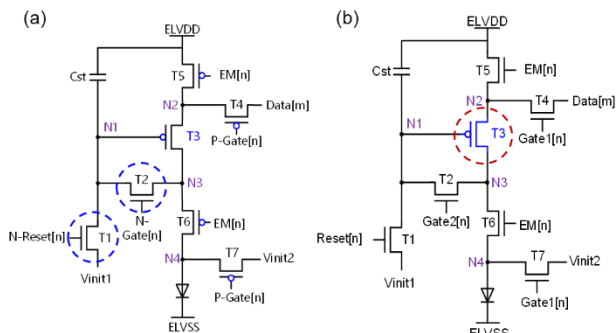


Figure 3. LTPO pixel circuits: (a) a conventional mass-production scheme, (b) the proposed scheme.

Current mass-produced LTPO pixel circuits typically employ 1-2 metal-oxide-semiconductor (MOS) TFTs, with the remaining being low-temperature polysilicon (LTPS) TFTs (Fig. 3a). We

propose a new LTPO circuit where only the driving TFT is an LTPS device, and all switching TFTs are MOS-based (Fig. 3b).

This design offers two key advantages. First, in conventional LTPO, TFTs are interconnected via conductivized LTPS, which has relatively low transmittance (Fig. 4). Our scheme uses conductivized MOS for interconnections, which exhibits a transmittance above 90%. Second, LTPS TFTs are highly sensitive to IR light [6], causing increases in off-state leakage current and threshold voltage shift. Shielding LTPS TFTs with light-blocking layers reduces the aperture ratio and transmittance. In contrast, MOS TFTs are inherently insensitive to IR light, eliminating this issue.

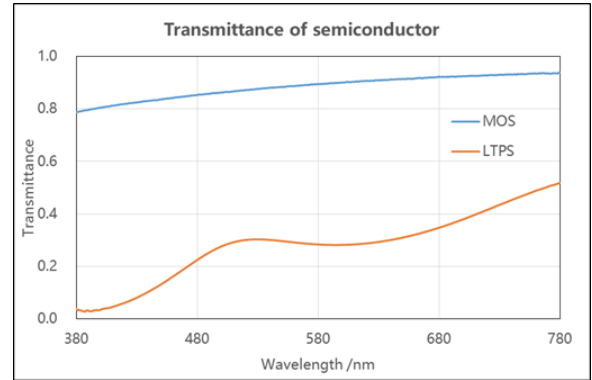


Figure 4. Transmittance comparison between LTPS and MOS films.

3.2 ITO Trace

Beyond TFT type, metal wiring is another major factor limiting transmittance. To improve the transmittance of the traces, we replaced the metal wiring of the gate lines, data lines, EM lines, and fixed-potential lines (initial lines) in the high-transmittance areas with a transparent conductive layer (ITO), as the transmittance of ITO exceeds 90%, while the main display area still uses metal wiring.

The use of ITO brings two problems. Firstly, ITO has poor hydrogen (H) blocking capability, which will affect the characteristics of MOS TFTs. Therefore, to prevent the channel from being affected during the metallization process, a small piece of metal is still retained above the channel region as a shield, as shown in Figure 5.

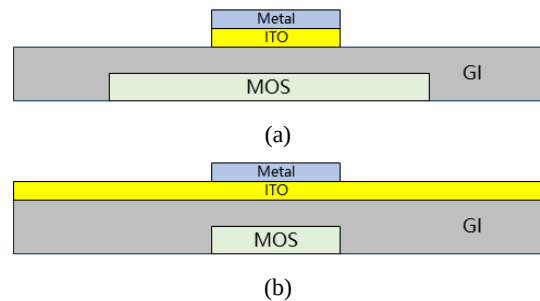


Figure 5. The cross-sectional structure of the MOS TFT (a) Y-direction, (b) X-direction.

Secondly, the resistance of ITO is much higher than that of metal, which will lead to greater resistance-capacitance (RC) delay. For this reason, we conducted simulations, and the specific results are

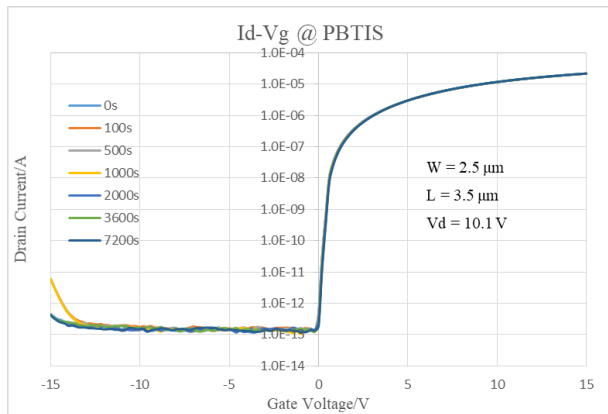
shown in Table 1 below. The results indicate that although ITO wiring slightly increases RC delay, this impact is still within a controllable range.

Table 1. The Impact of Transparent Wiring on RC Delay

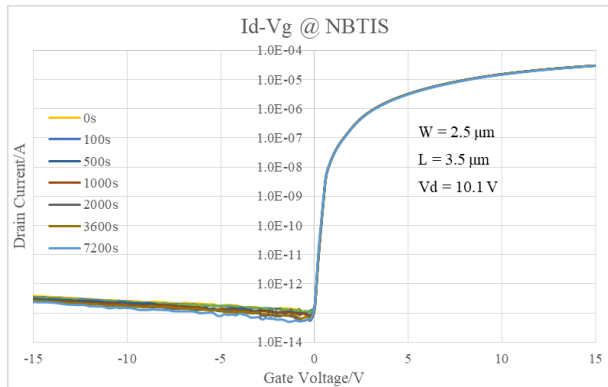
Signal type	RC Delay (μ s), Target<0.8 μ s	
	Main Display area	High Transmittance Area
Data	0.40	0.58
Gate	0.13	0.59
EM	0.28	0.64

3.3 High Mobility and Stability MOS TFT Device

Since ITO causes a certain impact on RC delay, to improve the charging rate and ensure a high refresh rate, we introduced a high-mobility MOS material with a mobility exceeding 25 cm²/Vs. Figure 6 shows the threshold voltage shift of these high-mobility oxide TFTs under positive and negative bias temperature illumination stress. Under conditions of ± 20 V gate bias, 70 °C, and 2500 μ W/cm² IR illumination for 2 hours, the TFTs exhibited excellent stability, with a V_{th} shift of -0.03 V under NBTIS and +0.11 V under PBTIS.



(a)



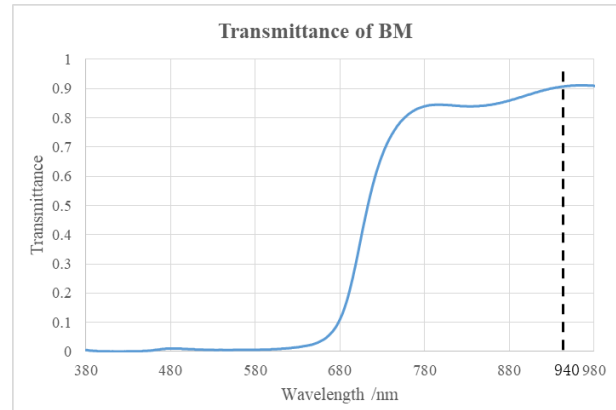
(b)

Figure 6. Stress reliability of high mobility MOS TFT (a) PBTIS, (b) NBTIS.

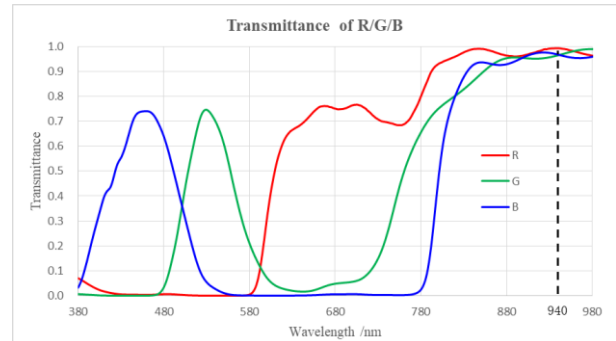
3.4 High IR-Transmittance COE

Although circular polarizers also have high infrared transmittance, we still choose color filters on the packaging layer (COE) as the anti reflection layer. Firstly, COE can reduce the power consumption of the screen. Secondly, compared to circular polarizers, the black matrix in COE can block ultraviolet and visible light, thereby reducing the impact of ambient light on TFT characteristics.

Unlike the traditional COE [10-11], the new COE technology tailored for UDFR requires high-transmittance black matrix (BM) and R/G/B materials, with their infrared transmittance exceeding 90%, as shown in Figure 7.



(a)



(b)

Figure 7. Transmittance spectrum of the new COE, (a) BM, (b) R/G/B.

OLEDs typically use semi-transparent cathodes to enhance light outcoupling. However, the transmittance of materials like Mg:Ag is significantly lower in the IR spectrum than in the visible range [7-8]. To improve IR transmittance, we implemented cathode patterning using a method consistent with a previous report [9-10].

By integrating the aforementioned innovations, we successfully developed a 6.0-inch FHD OLED screen with two high-transmittance areas. The structure of these areas is depicted in Fig. 8, achieving a final infrared transmittance of 29.64%.

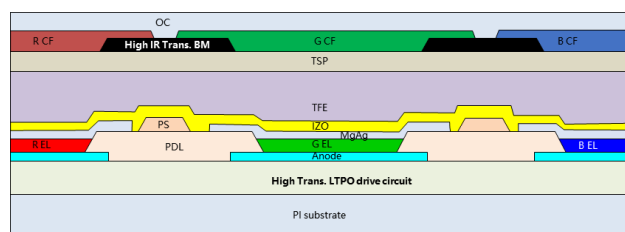


Figure 8. Layer structure of the display's high-transmittance area.

OLED Device Structure Design for Transparent Display .2022, SID Symposium Digest of Technical Papers, 2022, 53(S1): 194-196.

4. Conclusion

We have developed an UDFR-compatible OLED display using a new LTPO circuit with high-mobility, stable MOS TFTs, alongside transparent wiring and optimized COE. The prototype achieves 29.64% IR transmittance, demonstrating a viable path toward full-screen integration of face recognition in smartphones.

Acknowledgements

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